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Inequalities and optimisation

Level 3 Senior · Algebra

How to find and prove largest and smallest values using completing the square, AM-GM, the chain of means and extreme-case reasoning.

By the end you can: find maxima and minima by completing the square and by AM-GM with well-chosen splits; state, prove and use the QM-AM-GM-HM inequalities for two numbers; test extreme cases and check that a bound is actually attained.

Lesson

Start with a small problem. For positive x, what is the smallest value of $x + \frac{9}{x}$? Try a few values: $x = 1$ gives 10, $x = 2$ gives 6.5, $x = 3$ gives 6 and $x = 4$ gives 6.25. It looks as if the answer is 6, at $x = 3$. But a table of values proves nothing. Here is a proof in one line:

$$x + \frac{9}{x} - 6 = \frac{x^2 - 6x + 9}{x} = \frac{(x-3)^2}{x} \geq 0.$$

So $x + \frac{9}{x} \geq 6$ for every positive x, with equality only when $x = 3$. Two things happened there, and they are the two halves of every optimisation problem: a *bound* (the expression is never below 6) and an *example* showing that the bound is achieved (at $x = 3$). A bound without an example is half an answer, and an example without a bound is a guess.

Squares are never negative

Almost everything in this unit comes from one fact: a real square is never negative. The most direct use is completing the square. For $a > 0$,

$$ax^2 + bx + c = a\Big(x + \frac{b}{2a}\Big)^2 + c - \frac{b^2}{4a},$$

so the minimum is $c - \frac{b^2}{4a}$, reached at $x = -\frac{b}{2a}$. With several variables, complete the square one variable at a time, treating the others as constants, and use up the cross terms first.

Example 1. Find the minimum value of $x^2 + 2xy + 3y^2 - 4y + 5$ over all real x and y.

Solution.
A tempting idea is to minimise in each variable separately, for instance by setting $x = 0$ because $x^2 \geq 0$. That is wrong, because the term $2xy$ can be negative, so a non-zero x might help. The cross term couples the variables, and we must deal with it first.
Treat the expression as a quadratic in x with y fixed: $x^2 + 2xy = (x+y)^2 - y^2$. So the expression equals $(x + y)^2 + 2y^2 - 4y + 5$, and completing the square in y gives

$$(x + y)^2 + 2(y - 1)^2 + 3.$$

Both squares are non-negative, so the expression is at least 3. Equality needs $y = 1$ and $x = -y = -1$, and indeed $1 - 2 + 3 - 4 + 5 = 3$. The minimum is 3.

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AM-GM for two numbers

AM-GM. For non-negative a and b, $\frac{a+b}{2} \geq \sqrt{ab}$, with equality if and only if $a = b$.

Proof: $(\sqrt{a} - \sqrt{b})^2 \geq 0$ expands to $a + b - 2\sqrt{ab} \geq 0$, and equality holds exactly when $\sqrt{a} = \sqrt{b}$. The opening problem is AM-GM in disguise: $x + \frac{9}{x} \geq 2\sqrt{x \cdot \frac{9}{x}} = 6$. What made it work is that the product $x \cdot \frac{9}{x}$ is constant. That is the pattern to look for. AM-GM gives a constant lower bound for a sum when the product is fixed, and a constant upper bound for a product when the sum is fixed. Spotting which quantity is fixed is most of the work.

More terms, and the splitting trick

AM-GM for n numbers. For non-negative $a_1, \ldots, a_n$,

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n},$$

with equality if and only if all the a_i are equal.

Here is why. For four numbers, use the two-number version twice:

$$\frac{a+b+c+d}{4} = \frac{1}{2}\left(\frac{a+b}{2} + \frac{c+d}{2}\right) \geq \frac{\sqrt{ab} + \sqrt{cd}}{2} \geq \sqrt[4]{abcd}.$$

The same doubling gives 8, 16, and so on. For three numbers, let $m = \frac{a+b+c}{3}$ and apply the four-number version to a, b, c, m. Their mean is $\frac{3m+m}{4} = m$, so $m \geq \sqrt[4]{abcm}$, which gives $m^4 \geq abcm$ and so $m^3 \geq abc$ (if $m = 0$ everything is 0 anyway). The same padding idea handles every n, and following the equality cases through each step shows that equality needs all the numbers equal.

Example 2. For positive x, find the minimum value of $2x + \frac{27}{x^2}$.

Solution.
Two-term AM-GM gives $2x + \frac{27}{x^2} \geq 2\sqrt{\frac{54}{x}}$, but the right-hand side depends on x, so it is useless. A bound that moves is not a bound. We need pieces whose product is constant.
The trouble is that x appears to the power 1 in the first term and -2 in the second. Splitting $2x$ into two equal pieces $x + x$ fixes this, because $x \cdot x \cdot \frac{27}{x^2} = 27$. Three-term AM-GM now gives

$$x + x + \frac{27}{x^2} \geq 3\sqrt[3]{27} = 9.$$

Equality needs $x = \frac{27}{x^2}$, that is $x^3 = 27$, so $x = 3$. Check: $6 + 3 = 9$. The minimum is 9.
The lesson: split terms into equal pieces so that the product of all the pieces is constant, and make sure the equality case can actually happen.

Example 3. A cuboid has a square base. The total length of its twelve edges is 48 cm. Find the largest possible volume.

Solution.
Let the base have side x and the height be h. The edges give $8x + 4h = 48$, so $2x + h = 12$, and the volume is $V = x^2 h$. One option is to write $V = x^2(12 - 2x)$ and try values, but that only produces a guess. Instead notice that we have a fixed sum and want to bound a product, which is exactly what AM-GM does.

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The product is $x \cdot x \cdot h$, and the fixed sum $2x + h$ is $x + x + h$: a perfect match. So

$$4 = \frac{x + x + h}{3} \geq \sqrt[3]{x^2 h} = \sqrt[3]{V},$$

giving $V \leq 64$. Equality needs $x = h$, and with $2x + h = 12$ that means $x = h = 4$, a cube. The largest volume is $64\ \text{cm}^3$.
If the fixed sum had been $x + h$ instead, you would split it as $\frac{x}{2} + \frac{x}{2} + h$, so that the product of the pieces is $\frac{V}{4}$, still a constant multiple of V. Choosing the split is the art.

The chain of means

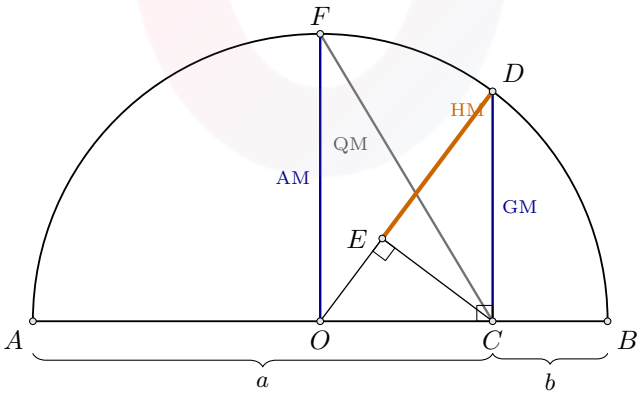
QM-AM-GM-HM. For positive a and b,

$$\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a + b},$$

with equality throughout if and only if $a = b$.

For the first inequality, square both sides: $\frac{a^2+b^2}{2} - \frac{(a+b)^2}{4} = \frac{(a-b)^2}{4} \geq 0$. The middle one is AM-GM. For the last, note that $\frac{2ab}{a+b} = \frac{ab}{(a+b)/2}$, so the ratio of the geometric mean to the harmonic mean equals the ratio of the arithmetic mean to the geometric mean, which is at least 1.

There is a lovely picture. Take a semicircle on a diameter AB of length $a + b$, with centre O, and mark C on AB with $AC = a$ and $CB = b$. Then the radius OF perpendicular to AB is the arithmetic mean. The perpendicular CD to AB is the geometric mean, since $CD^2 = AC \cdot CB$ by similar triangles. Dropping the perpendicular CE to OD gives $DE \cdot DO = DC^2$, so DE is the harmonic mean. Finally $OC = \frac{|a-b|}{2}$, so Pythagoras gives $CF = \sqrt{\frac{a^2+b^2}{2}}$.



Each inequality is now "a hypotenuse is longer than a leg": $CF \geq OF$, $OD \geq CD$ and $CD \geq DE$. In contests, QM-AM is the tool when you know a sum of squares and want to bound a sum (or the reverse), and the harmonic mean turns up whenever rates are averaged.

Extreme cases and bounding

Not every optimisation problem is AM-GM. Two habits cover most of the rest. First, *test the extreme cases before you prove anything*: make the variables equal, make one of them as large as possible, set some to zero. In symmetric problems one end of the range is often at the symmetric point and the

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other at a boundary. These tests tell you what answer to aim for, and so which inequality you need. Second, *if you bound pieces separately, check that the bounds are reached at the same point.* If you prove $A \le 3$ and $B \le 1$, then $A + B \le 4$, but that is only the maximum if both equalities can hold together.

Example 4. Non-negative reals a, b, c satisfy $a + b + c = 3$. Find the largest and smallest possible values of $S = ab + bc + ca + abc$.

Solution.
Test some cases first. At $a = b = c = 1$ we get $S = 3 + 1 = 4$. At $(3, 0, 0)$ we get $S = 0$. At $(\frac{3}{2}, \frac{3}{2}, 0)$ we get $S = \frac{9}{4}$. So we guess the maximum is 4 at the symmetric point and the minimum is 0 at the boundary.
The minimum is immediate: every term of S is non-negative, so $S \ge 0$, and $(3, 0, 0)$ gives 0.
For the maximum, bound the two parts separately. Since

$$(a+b+c)^2 - 3(ab+bc+ca) = a^2+b^2+c^2-ab-bc-ca = \tfrac{1}{2}\big((a-b)^2+(b-c)^2+(c-a)^2\big) \ge 0,$$

we get $ab + bc + ca \le \frac{9}{3} = 3$. By AM-GM, $abc \le \left(\frac{a+b+c}{3}\right)^3 = 1$. So $S \le 4$, and both bounds are equalities at $a = b = c = 1$, where $S = 4$. The largest value is 4 and the smallest is 0.
Splitting worked only because both pieces peak at the same point. When they peak at different points, separate bounds give a true statement that is not the answer.

Common traps.

- Equality must be achievable. For $x \ge 5$ it is true that $x + \frac{9}{x} \ge 6$, but equality would need $x = 3$, which is not allowed. The real minimum is $5 + \frac{9}{5} = \frac{34}{5}$, because the expression increases for $x \ge 3$.
- AM-GM needs non-negative numbers. The claim $x + \frac{1}{x} \ge 2$ is false for negative x.
- A bound that depends on the variable, such as $2\sqrt{54/x}$ in Example 2, is not a bound.
- Separate bounds must be attained at the same point, as in Example 4.
- Check that a maximum really exists. On $x > 0$, the expression $1 - \frac{1}{x}$ is always below 1 and gets as close as you like, but never reaches it, so it has no maximum.

How to spot this in a contest

The phrase "smallest possible value" or "largest possible value" is the obvious signal, and then the shape of the expression chooses the tool. A sum of terms whose product is constant, or a product with a fixed sum, wants AM-GM, after splitting if needed. A quadratic in one or more variables wants completing the square. A symmetric constraint means testing equal values and boundary values first. Integer variables usually mean an extreme-case argument: give the most important variable its best possible value and check nothing beats it.

On a multiple-choice paper, finding the equality case and evaluating there often identifies the answer quickly. Keep the proof in mind anyway, because the same problems reappear at olympiad level where only the proof scores.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ For positive x, find the smallest value of $9x + \frac{16}{x}$.

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2. ⋆⋆ Real numbers x and y satisfy $x + 2y = 10$. Find the smallest possible value of $x^2 + y^2$.

3. ⋆⋆ Find the largest value of $\frac{x}{x^2+16}$ over all real x.

4. ⋆⋆⋆ The equation $x^4 - 12x^3 + ax^2 + bx + 81 = 0$ has four positive real roots, not necessarily distinct. Find $a - b$.

5. ⋆⋆⋆ Positive reals a, b and c satisfy $a + b + c = 1$. Find the largest possible value of ab^2c^3.

6. ⋆⋆⋆⋆ Real numbers x, y and z satisfy $x + y + z = 0$ and $x^2 + y^2 + z^2 = 6$. Find the largest possible value of xyz.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. UKMT SMC 2020 Question 19 ⋆
2. AMC 12A 2021 Problem 7 ⋆⋆
3. UKMT SMC 2017 Question 18 ⋆⋆
4. AHSME 1965 Problem 34 ⋆⋆
5. UKMT SMC 2021 Question 23 ⋆⋆⋆
6. AHSME 1979 Problem 29 ⋆⋆⋆
7. AMC 12A 2003 Problem 24 ⋆⋆⋆
8. UKMT SMC 2022 Question 24 ⋆⋆⋆
9. AHSME 1982 Problem 29 ⋆⋆⋆⋆
10. AMC 12A 2016 Problem 24 ⋆⋆⋆⋆

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Hints

1. The product of the two terms does not depend on x.
2. Use the constraint to remove x, then complete the square.
3. Only positive x can give a positive value. For those, divide the top and bottom by x.
4. Use Vieta to find the sum and the product of the roots, then compare their arithmetic and geometric means.
5. Split b into two equal pieces and c into three, so that you have six pieces with a fixed sum.
6. Fix z. Express $x+y$ and xy in terms of z, and ask what it takes for x and y to be real.

Solutions

Problem 1

Answer: 24

By AM-GM, $9x + \frac{16}{x} \geq 2\sqrt{9x \cdot \frac{16}{x}} = 2\sqrt{144} = 24$. Equality holds when $9x = \frac{16}{x}$, that is $x = \frac{4}{3}$, and then the expression is $12 + 12 = 24$. So the minimum is 24.

Problem 2

Answer: 20

Substituting $x = 10 - 2y$ gives $x^2 + y^2 = (10-2y)^2 + y^2 = 5y^2 - 40y + 100 = 5(y-4)^2 + 20$. This is at least 20, with equality when $y = 4$ and $x = 2$. So the minimum is 20. Geometrically, $x^2 + y^2$ is the squared distance from the origin to a point of the line $x + 2y = 10$, and the closest point is $(2, 4)$.

Problem 3

Answer: 1/8

For $x \leq 0$ the fraction is not positive, so the maximum, if it exists, comes from $x > 0$. For $x > 0$ we can write $\frac{x}{x^2+16} = \frac{1}{x+\frac{16}{x}}$. By AM-GM, $x + \frac{16}{x} \geq 2\sqrt{16} = 8$, with equality at $x = 4$. So the fraction is at most $\frac{1}{8}$, and it equals $\frac{1}{8}$ at $x = 4$. The largest value is $\frac{1}{8}$.

Problem 4

Answer: 162

Let the roots be $r_1, r_2, r_3, r_4 > 0$. By Vieta's formulas, $r_1 + r_2 + r_3 + r_4 = 12$ and $r_1 r_2 r_3 r_4 = 81$. So the arithmetic mean of the roots is 3, and their geometric mean is $\sqrt[4]{81} = 3$ as well. AM-GM says that the arithmetic mean is at least the geometric mean, with equality only when all the numbers are equal. Here equality holds, so every root equals 3.

Hence the polynomial is $(x-3)^4 = x^4 - 12x^3 + 54x^2 - 108x + 81$, giving $a = 54$, $b = -108$ and $a - b = 162$.

Problem 5

Answer: 1/432

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Write $1 = a + \frac{b}{2} + \frac{b}{2} + \frac{c}{3} + \frac{c}{3} + \frac{c}{3}$ and apply AM-GM to these six positive numbers:

$$\frac{1}{6} \geq \sqrt[6]{a \cdot \frac{b^2}{4} \cdot \frac{c^3}{27}}.$$

Raising both sides to the sixth power gives $\frac{ab^2c^3}{108} \leq \frac{1}{6^6}$, so $ab^2c^3 \leq \frac{108}{46656} = \frac{1}{432}$.

Equality needs $a = \frac{b}{2} = \frac{c}{3}$, and together with $a + b + c = 1$ this gives $a = \frac{1}{6}$, $b = \frac{1}{3}$ and $c = \frac{1}{2}$. There $ab^2c^3 = \frac{1}{6} \cdot \frac{1}{9} \cdot \frac{1}{8} = \frac{1}{432}$, so the largest value is $\frac{1}{432}$. The split was chosen so that the six pieces can all be equal at the same time as the constraint holds, which is exactly what makes the bound attainable.

Problem 6

Answer: 2

Squaring $x + y + z = 0$ gives $x^2 + y^2 + z^2 + 2(xy + yz + zx) = 0$, so $xy + yz + zx = -3$. Now fix z. Then $x + y = -z$ and $xy = -3 - z(x + y) = z^2 - 3$. So x and y are the roots of $t^2 + zt + z^2 - 3 = 0$, and they are real exactly when the discriminant is non-negative: $z^2 - 4(z^2 - 3) \geq 0$, that is $z^2 \leq 4$. Conversely, any z with $-2 \leq z \leq 2$ gives real x and y satisfying both conditions.

Now $xyz = z(z^2 - 3) = z^3 - 3z$, and we want its largest value for $-2 \leq z \leq 2$. We have

$$2 - (z^3 - 3z) = -(z - 2)(z + 1)^2 = (2 - z)(z + 1)^2 \geq 0$$

for $z \leq 2$, so $xyz \leq 2$. Equality holds at $z = 2$, where $x = y = -1$, and these values satisfy both conditions, with $xyz = 2$. So the largest value is 2.

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